STAT 464 Assignment 2 Question 2

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Set up time series

```
Z_t <- rnorm(100, 0, 2)
t <- c(1:100)

X_t <- 0.1*t + Z_t
```

ACF Plotting

```
h <- c(0:50)

acf(X_t, lag.max = 50,
    main = expression(paste("Sample ACF of ", X[t])),
    col = "darkgray", lwd = 2)
```

A2 Q2

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$X_t = 0.1t + Z_t$

$E[X_t] = E[0.1t] + E[Z_t]$ & Z_t has $\mu = 0$

$= 0.1t$

$\therefore X_t$ is non-stationary b/c the mean relies on t.

However, we can easily show that the ACVF and ACF do not rely on t, so we may define them

$\gamma_X(h) = Cov(X_t, X_{t+h})$

$= Cov(0.1t + Z_t, 0.1(t+h) + Z_{t+h})$

Clearly, we see that the first terms overlap only when $h = 0$. Likewise for the WN terms.
So, $\gamma_X(h) = 0$ for $h \neq 0$.
As for when $h = 0$, we see that the first terms may be treated as constants since t is not distributed and has 0 variance.

$\gamma_X(0) = Cov(Z_t, Z_{t+0})$

$= Cov(Z_t, Z_t)$

$= 2$

$\therefore \gamma_X(h)$ is well-defined since it doesn't rely on t, and

$$\gamma_X(h) = \begin{cases} 2 & \text{for } h = 0 \\ 0 & \text{for } h \neq 0 \end{cases}$$

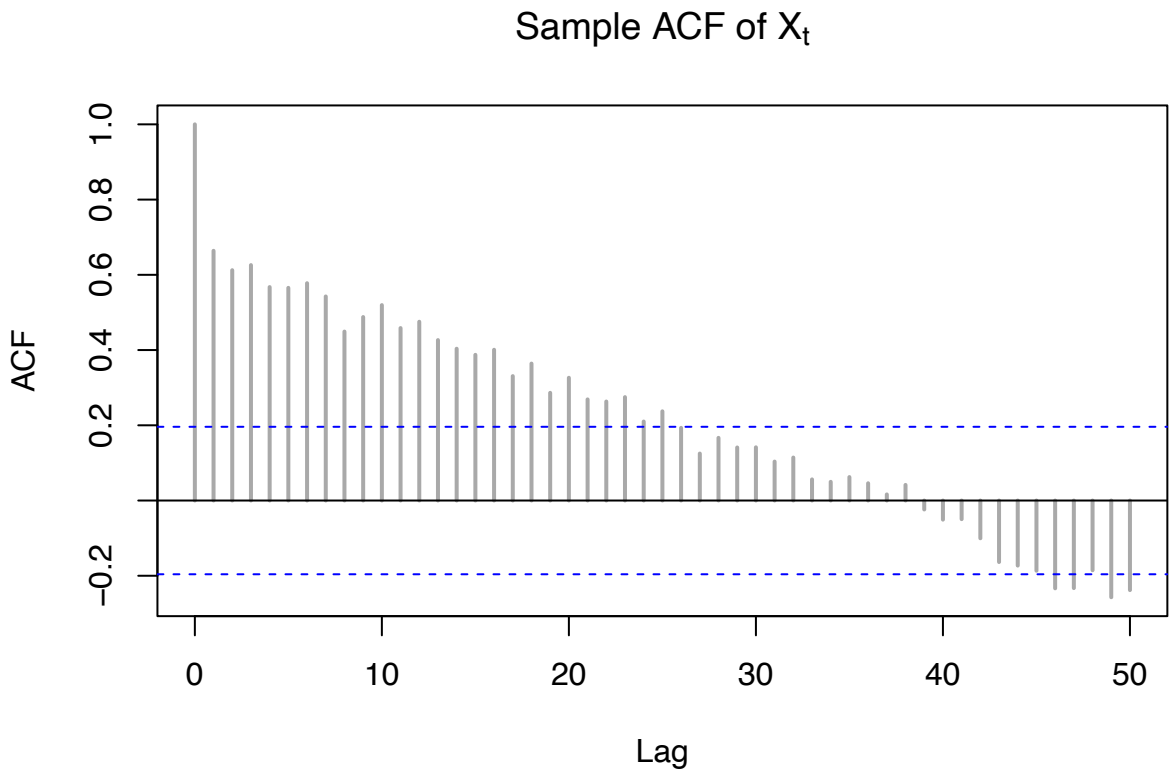
For the ACF

$$\rho_X(h) = \frac{\gamma_X(h)}{\gamma_X(0)}$$

$$= \begin{cases} \frac{2}{2} & \text{for } h = 0 \\ \frac{0}{2} & \text{for } h \neq 0 \end{cases}$$

$\therefore$ The true ACF, well-defined since it doesn't depend on t, is:

$$\rho_X(h) = \begin{cases} 1 & \text{for } h = 0 \\ 0 & \text{for } h \neq 0 \end{cases}$$



This ACF has the form it does by attempting to average towards 0 for the ACF when the lag `h != 0`. At `h = 0`, clearly we see that the ACF = 1 just like with the true/theoretical ACF computed. However, at `h != 0` it isn't exactly ACF = 0 like the theoretical formula, but trends towards it and likely will have some more negative lags `h > 50` to produce an average ~= 0.